JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 92, NO. B5, PAGES 3580-3596, APRIL 10, 1987

EASTERN ASIA IN THE CRETACEOUS: NEW PALEOMAGNETIC DATA FROM SOUTH KOREA AND A NEW LOOK AT CHINESE AND JAPANESE DATA

Lee Gidong[1], Jean Besse, and Vincent Courtillot

Laboratoire de Paléomagnétisme et Géodynamique, Institut de Physique du Globe de Paris and UER Sciences Physiques de la Terre, Université Paris 7

Raymond Montigny

Institut de Physique du Globe de Strasbourg, Université Pasteur

Abstract. Lower Cretaceous continental sediments and volcanic rocks have been sampled in the Gyeongsang basin (South Korea). A detailed analysis of 160 specimens reveals two stable components of magnetization, a postfolding, low-temperature (LT) component with normal polarity and a stable, high-temperature (HT) component with both normal and reversed polarities. The blocking temperature of the HT component ranges from 540°C up to the Néel temperature of hematite. This component predates the folding of the series and yields a pole at $\lambda = 68^{\circ}$N, $\phi = 205^{\circ}$E, $A_{95} = 6^{\circ}$. The prefolding magnetization indicates that South Korea stood at a latitude nearly identical to its present-day latitude in the Lower Cretaceous. Comparison of Korean data with (revised) paleomagnetic data from North and South China and Japan shows that all these blocks of Eastern Asia occupied the same positions in term of latitude and were probably parts of a single craton since at least the Lower Cretaceous. The paleomagnetic directions of the Sino-Korean block show large discrepancies with respect to the expected directions of the "stable" Eurasian (Siberian) block. An interpretation involving significant N-S motion (~1000 km) between either North and South China or between North China and Siberia appears to contradict geological data. A preferred interpretation is that the Siberian data are in serious error and that the combined China/Korea Lower Cretaceous pole provides a valid estimate for the Eurasian plate as a whole.

1. Introduction

The eastern part of Eurasia comprises three major cratonic blocks: Siberia, North China (often referred to as the Sino-Korean block), and South China (also known as the Yangtze platform). The three blocks are separated by major suture zones, considered to be of late Paleozoic age, and certainly not later than early Jurassic (for instance, Klimetz [1983] and Chinese Academy of Sciences [1982]; see Figure 1). Hence most authors consider this part of the Asian continent to have remained essentially stable since the Late Jurassic or Early Cretaceous. In this framework, Korea occupies an interesting position. It is generally considered to be a part of the stable North China block and is not thought to have undergone any significant motion with respect to the rest of North China and Siberia in the last 150 m.y. However, significant tectonic activity has occurred since that time around the southern and eastern margins of the Asian continent. On one hand, continued subduction of the Kula and Pacific plates to the east climaxed with the Late Cretaceous-Paleocene collision of the Sikhote Alin block and Cenozoic opening of the Japan Sea. On the other hand, India collided with Asia in the Eocene. Molnar and Tapponnier [1975] and Tapponnier et al. [1986] have shown that this collision resulted in large-scale intracontinental deformation, extending with decreasing amplitude to the north and east throughout much of Asia. In the light of this, the traditional view of a stable Mesozoic assemblage of China and Korea deserves further testing.

We collected continental sediments and volcanic samples of Lower Cretaceous age along two sections in the Gyeongsang basin of South Korea (Figures 1-3). The paleomagnetic analysis of these samples yields a Lower Cretaceous pole which is then compared to results available for North Korea, Japan, North and South China, and the rest of Eurasia.

2. An Overview of Korean Geology

General setting. The Korean peninsula is bounded to the east by the Sea of Japan (Figure 1), a marginal sea thought to have opened in the Cenozoic [Kobayashi, 1983; Kobayashi and Isezaki, 1976], and to the west by the Yellow Sea, which is underlain by a continental crust [e.g., Ma et al., 1982]. Going from North China to the NW to South Japan to the SE, Korea exhibits NE-SW trending tectonic boundaries, many of them still not well known. To the north, the Korea block is separated from the North China block by part of the very extensive NNE-trending Tancheng Lujiang (or Tanlu) system of strike-slip faults and basins. This major left-lateral fault system extends over 2500 km from the Yangtze River to north of the Sakhalin block [Xu, 1980]. Farther to the southeast, the peninsula is divided by the Shugareong zone, considered to be a Paleozoic suture [Chinese Academy of Sciences, 1982]. The trend and exact

[1] Permanently at Korean Institute of Energy and Resources, Seoul, Republic of Korea.

Copyright 1987 by the American Geophysical Union.

Paper number 5B5730.
0148-0227/87/005B-5730$05.00

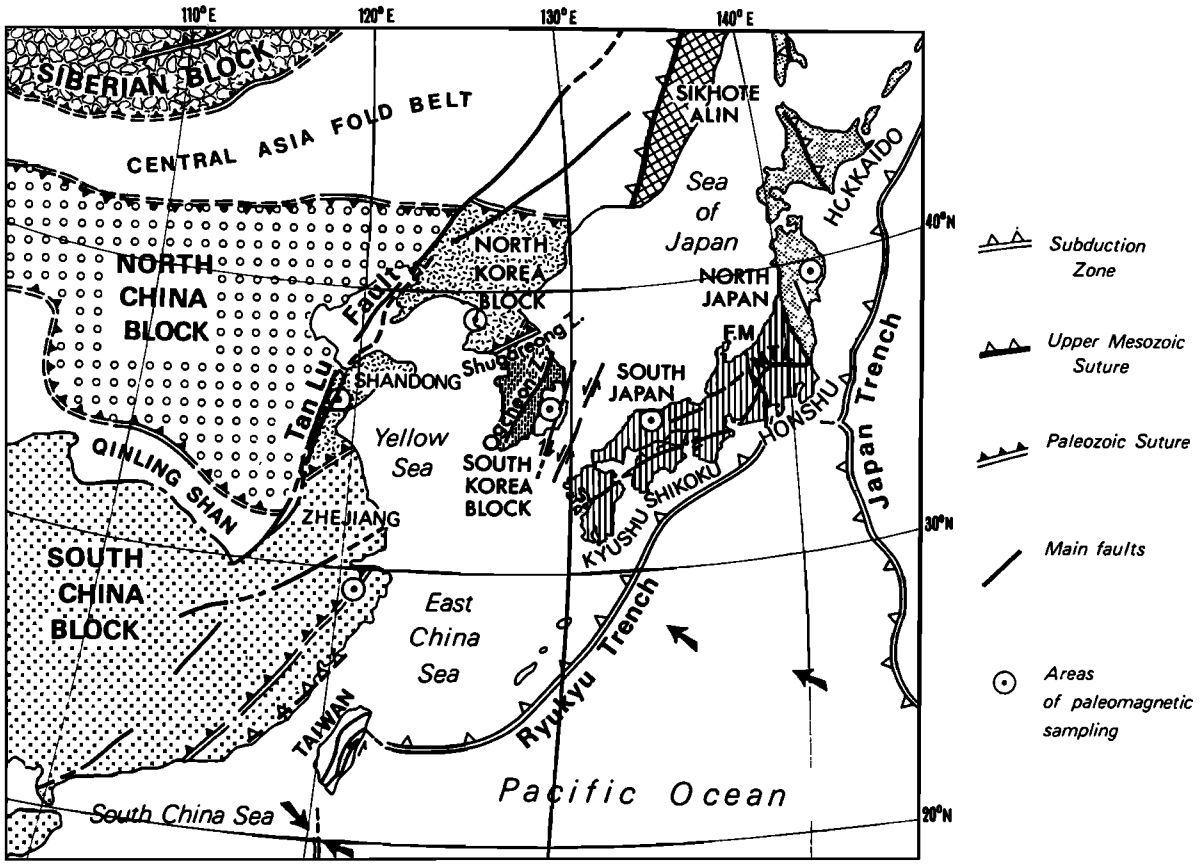


Fig. 1. Schematic tectonic map of Southeastern Asia showing the major blocks and their boundaries. Major sites of paleomagnetic information are shown as circles.

location of this suture are not immediately clear on the simplified geological map of Korea (Figure 2). Far more obvious is the NNE trending Ogcheon zone farther south (Figures 1 and 2), mapped as a strike-slip fault by Chinese Academy of Sciences [1982] and in which strong N60° folding occurs. Finally, the southeastern coast of Korea and the Tsushima (or Korea) strait are cut by recent NNE trending right-lateral strike-slip faults. Active subduction takes place to the south and east, in the western Pacific; the closest subduction occurs at the Ryukyu trench about 500 km to the southeast.

The geological map of Korea (Figure 2 [see Sang and Hee, 1983]) shows that most of the peninsula is composed of Precambrian to Triassic formations, except for the Cretaceous Gyeongsang basin to the southeast (Figures 2 and 3). These are intruded by granites and granodiorites with ages ranging from Triassic to Tertiary. Triassic granites are found only north of the Shugareong zone, and Cretaceous granites and volcanics are found mostly across the Ogcheon zone and to the south of it. The Jurassic granites are the most abundant and are dominant in the central third of Korea, between the Chugareong and Ogcheon zones, although they also occur both north and south of these zones.

Gyeongsang basin: During the Cretaceous, the southeastern part of the Korean peninsula was filled by more than 10 km of sediments which rest unconformably over Precambrian gneisses and Jurassic granites (Figure 2). The series is gently folded, and the dip is small (10°-20°). In its western part, the series is actually monoclinal, dipping mostly toward the ESE (Figure 4). To the east, the Cretaceous series is uncomformably overlain by Tertiary marine sediments and volcanics of Miocene to Quaternary age. The Gyeongsang basin is divided by Korean geologists into several subunits [see Chang, 1975]: the Nagdong trough, the Silla subbasin, the Yeongyang subbasin, and the Dopyeong block (Figure 3). The series consists of continental sediments, with interbedded basaltic flows corresponding to moderate volcanic activity. The different units of the Gyeongsang basin display important lateral facies variations [Chang, 1975]. Since they are made of conglomeratic or detrital material, age determinations are difficult. An analysis of pollens found in the coaly levels of the series sampled for the present paleomagnetic study has been performed in the palynologic laboratories of Total Petroleum Company in France: Unfortunately, all it provides is an age window from Upper Jurassic to Upper Aptian. A recent study, based on new determinations of Charophytes, gives a Barremian to Aptian age (K. H. Chang, personnal communication, 1984), that is roughly between 125 and 110 m.y. Yu [1971] indeed found an Aptian age for the two uppermost formations of the Nagdong series, also based on Charophyte determinations.

The radiometric age determinations for

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

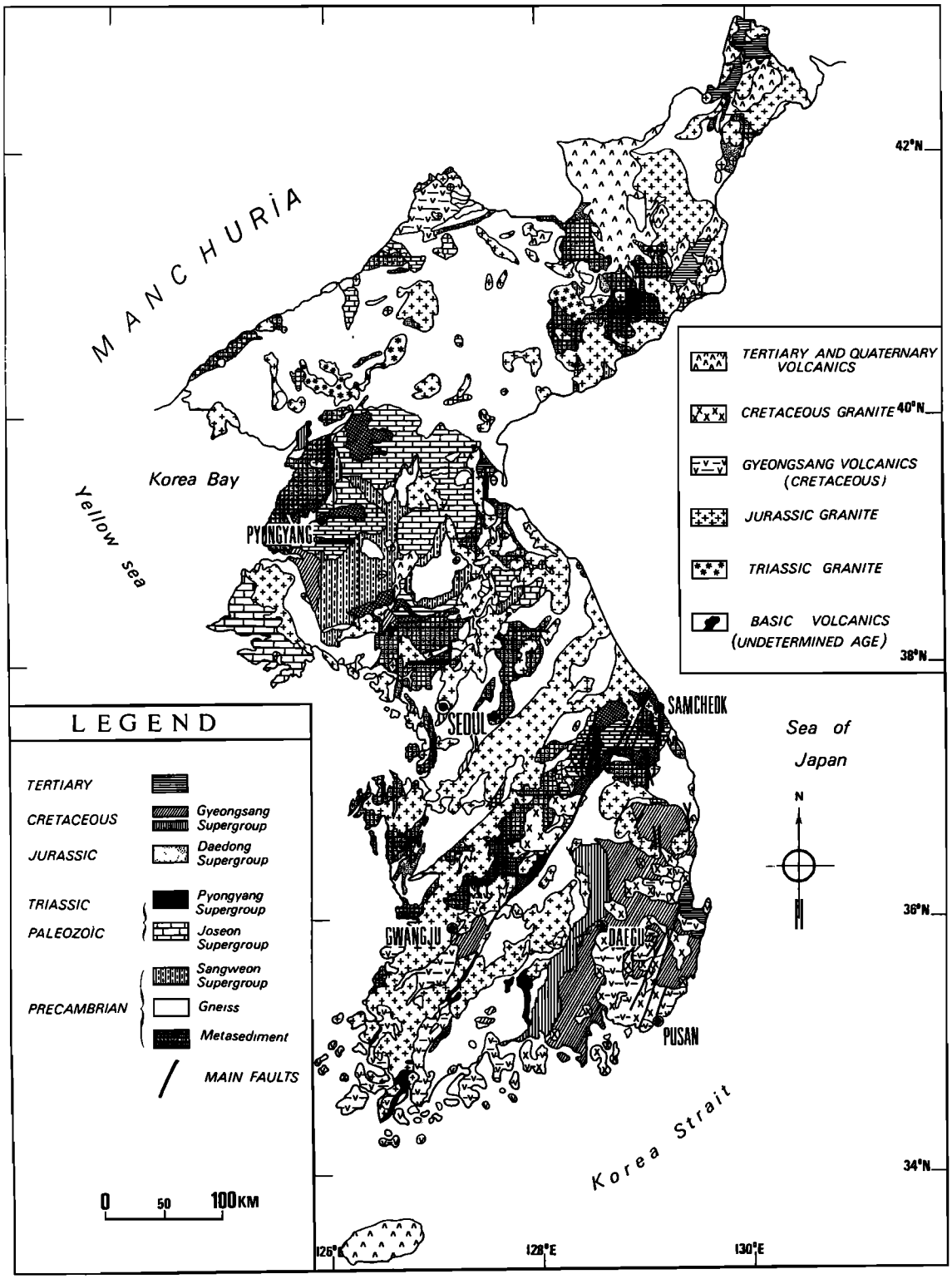


Fig. 2. Simplified geological map of Korea.

igneous rocks are somewhat conflicting: Intrusive granites give an upper age boundary going from 68 to 88 m.y. [Kim, 1982; Suh and Choo, 1971]. Otofuji et al. [1983] propose an age of 68 ± 3 m.y. (K/Ar) for the Hagbong diabase. We have performed a new K/Ar determination which gives 70.5 ± 2.5 m.y. (diabase sample COR 84-3A), in agreement with that of Otofuji et al (1983). This age is much younger than the age of the host rocks (Figure 3) and can be explained in several ways: Either the volcanic flows are interstratified in the sediments and the age would have suffered from rejuvenation due for instance to weathering or metamorphism, or the

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

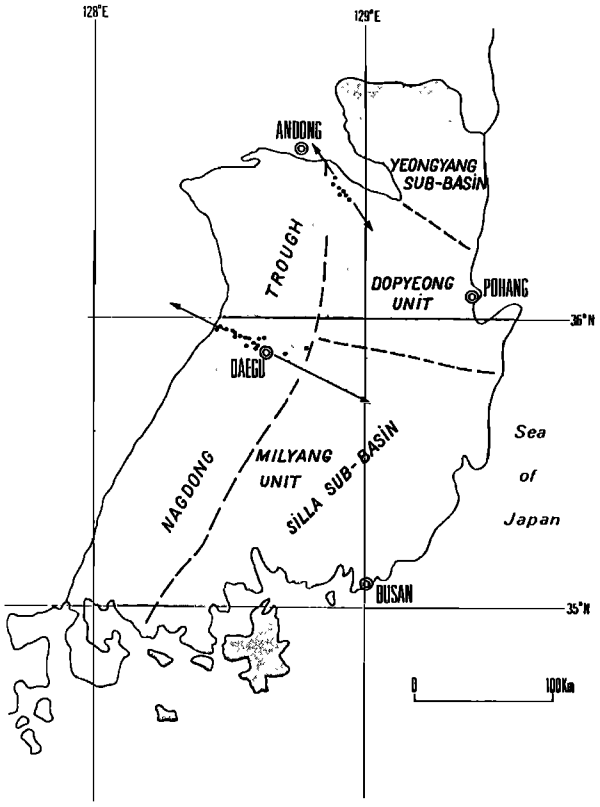


Fig. 3. Map of paleomagnetic sampling sites in the Gyeongsang basin (Daegu and Andong areas). Dashed lines correspond to the subdivision into paleogeographic provinces proposed by Korean geologists.

volcanics are a younger sill. In any case, two additional age determinations from andesites that cut across the youngest part of the Gyeongsang series (Figure 4, to the ESE), at 88 ± 4 m.y. (sample COR 84-4C) and 90 ± 4 m.y.(sample COR 84-4B), give a minimum age which does not contradict the paleontological determination.

3. Paleomagnetic Study

Sampling and procedure. We have collected a total of 78 blocks from 25 sites (Figure 3). The blocks were oriented mostly with a magnetic compass and with a sun compass when possible. In the laboratory, two specimens on average were cored from each block. Specimens were demagnetized by step heating followed by cooling in approximately field-free space (<20 nT) or by use of increasing alternating fields (af). In addition, some of the samples were chemically demagnetized, using increasing concentrations of hydrochloric acid. Magnetic measurements were carried out either on a spinner magnetometer (Schönstedt GSD-1) or on a two-component cryogenic magnetometer (CTF). Most samples had weak multicomponent magnetizations which were determined using a least squares component analysis. A detailed account of the measurements is given by Lee [1984].

Daegu section (measurements). Eighteen sites were sampled along a WNW-ESE section in the Nagdong trough (Figures 3 and 4). As discussed above, the age of the series probably extends from Barremian to Aptian. Fifteen sites provided reliable results (sites 3-18, see Figure 4) which are illustrated by vector demagnetization diagrams (Figures 5 and 6). Most diagrams display two and more typically three more or less overlapping components. A first, softer component is usually eliminated by 5-10 mT (af) or 120°-250°C (thermal). This soft component has a random direction and is not considered further. A second component A, often the dominant one, is isolated between 120°-250°C and 350°-650°C, with unblocking temperatures in both the magnetite and hematite range. Both minerals are found, depending on the site, as shown by joint consideration of the af and thermal diagrams. A third component B with a high unblocking temperature is found in many samples, sometimes only above 650°C. The diagrams show this component going through the origin, within experimental accuracy. This component often amounts to less than 10% of the original remanent magnetization (NRM) and was computed using least squares ana-

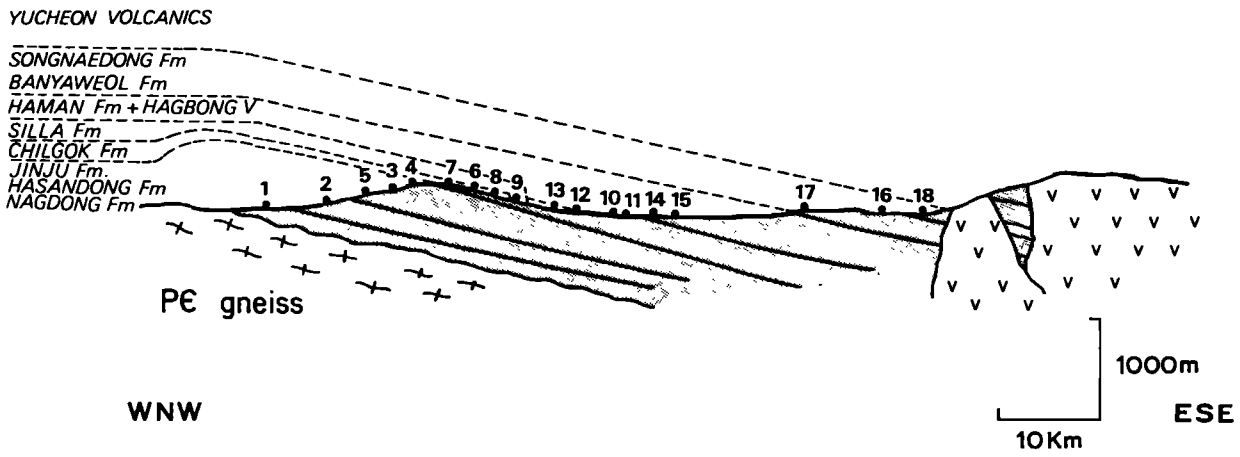


Fig. 4. Cross section through Daegu in the Gyeongsang basin and location of paleomagnetic sampling sites. The shaded zone represents continental sediments, and the v pattern stands for post-Late Cretaceous calc-alkaline intrusives.

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

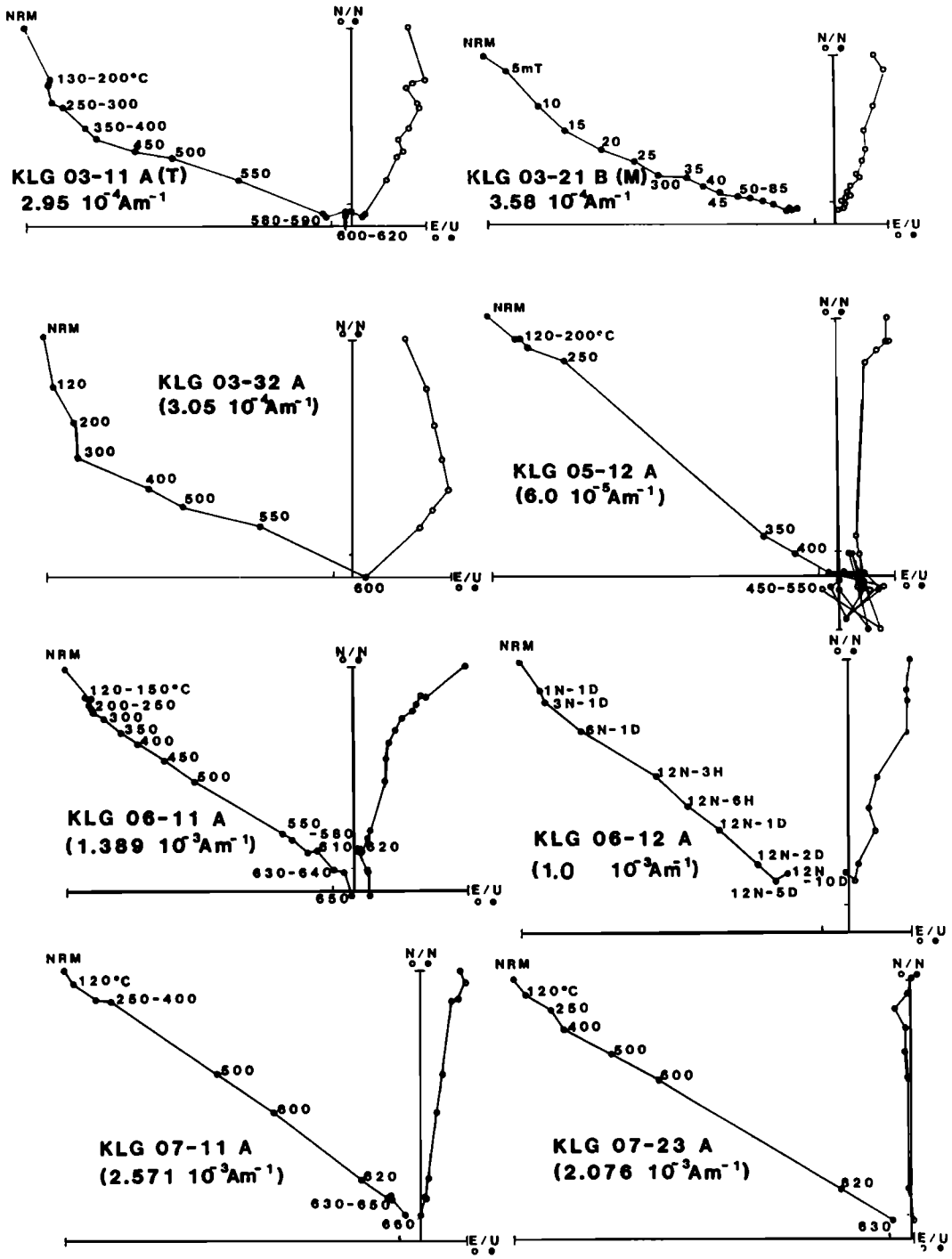


Fig. 5. Vector demagnetization diagrams of specimens from the Daegu section. Solid symbols are the projections of the magnetic vector onto the north-south vertical plane, open symbols onto the horizontal plane. The intensity corresponding to tick marks on the axes is given near the specimen code name.

lysis method on the last five steps on average, and sometimes is not seen at all. Special care was taken in isolating it, as it will be proposed that B represents the primary magnetization. A component with direction intermediate between A and B is sometimes found and may either be real or may correspond to overlapping unblocking temperatures (or coercivity spectra) of A and B. Details on demagnetization behavior is given below for groups of sites (in stratigraphic order of formations), with particular emphasis on stability indicators such as the presence of reversals or a positive fold test when available. Results are summarized in Table 1.

Sites 1 and 2 (Nagdong Hasandong Formation;

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

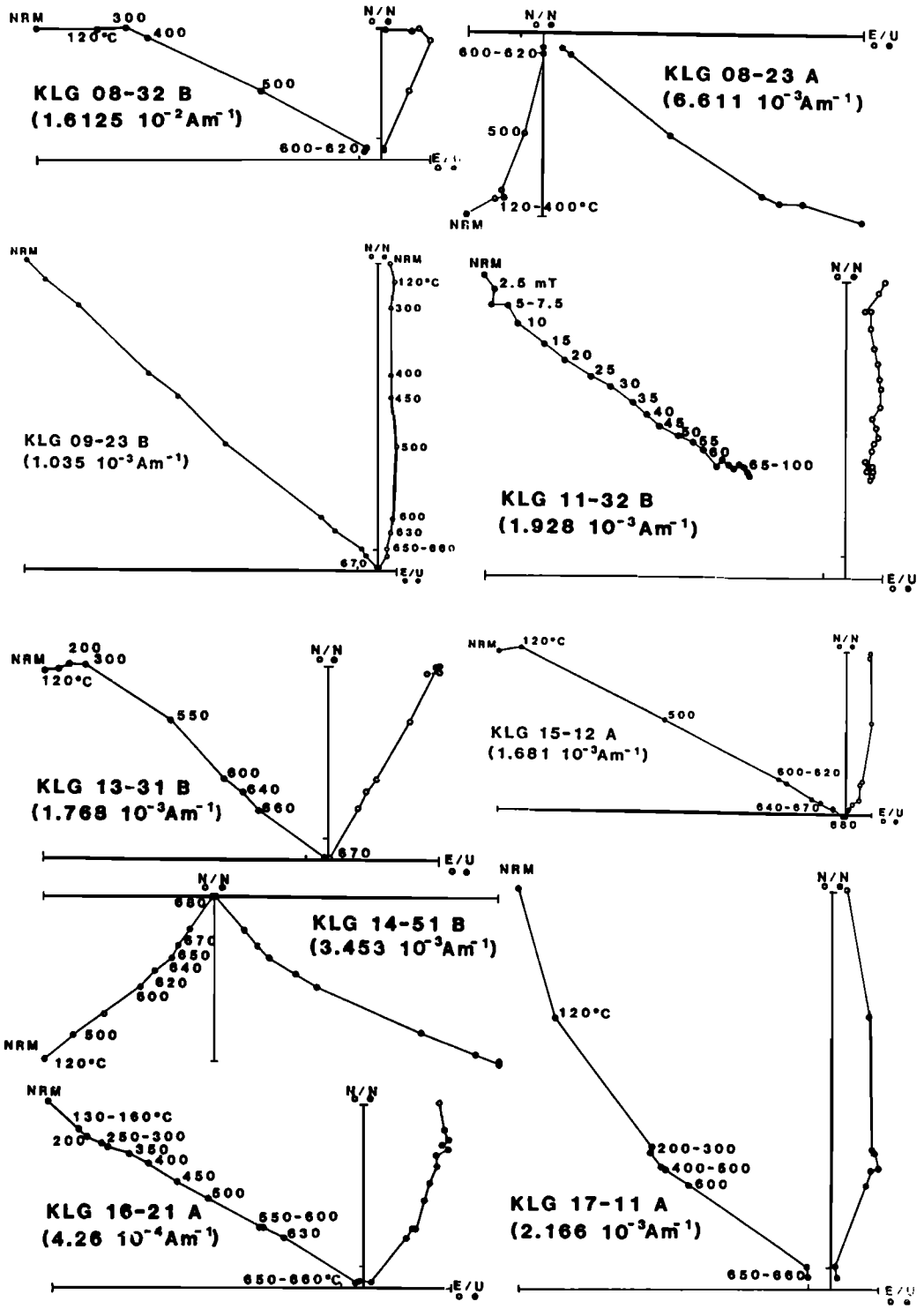


Fig. 6. Same as Figure 5.

sandstones): The thermal demagnetization diagrams display an erratic behavior of the directions. The two sites have been discarded.

Sites 3-5 (Jinju formation; gray sandstones; Figure 5): Site 3 provides consistent results with both thermal and af treatments. Component B is isolated between 400° and 600° C (specimens 3-11A, 3-32A, Figure 5). Thermal treatment is more efficient than af treatment in separating components A and B (specimen 3-21B, Figure 5). In many specimens from sites 4 and 5, only the lower-temperature component A is found, and an erratic behavior occurs above 450° C (e.g., specimen 5-12A, Figure 5). Only four specimens display the high-temperature component. The carrier of magnetizations appears to be a form of magnetite in these sites.

Sites 6-8 (Chilgok formation; mudstones and

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Directions in Geographic (Dg, Ig) and Stratigraphic (Ds, Is) Coordinates of the High-Temperature Components of the Daegu Section

Site	Nb	n	BA	Dg	Ig	Ds	Is	α_{95}	λ(N)	ϕ(E)
01	5	5		unstable						
02	3	5		unstable						
03*	3	12	16,20 NE	23.9	66.0	44.8	50.3	4.6	52.6	212.8
04-05*	4	4	49,22 SE	7.1	58.2	22.5	64.9	16	70.0	178.5
06*	3	11	54,18 SE	8.3	56.2	36.5	65.9	5.4	60.6	182.5
07*	4	8	60,28 SE	13.8	43.6	44.6	58.7	7.3	54.9	199.1
08+*	4	9	69,20 SE	15.1	53.5	42.3	66.9	6.5	56.6	181.1
08-*	2	4		198.4	-52.9	225.8	-65.4	4.9	54.5	185.0
09*	3	12	84,18 SE	13.1	40.2	27.4	55.3	8.7	67.9	210.7
11	3	17	67,10 SE	16.1	51.6	26.1	58.8	4.4		
12	4	17	30,14 SE	2.3	53.7	22.6	58	1.7		
13	4	14	29,24 SE	10.4	55.6	46.8	55.7	5.6		
MV*	11	48		9.7	53.8	32	58	11	64.5	202.7
14+*	3	5	54,10 SE	10.4	38.5	17.2	45	8.3	72.6	246.3
14-*	2	4		214.0	-35.2	221.0	-38.1	5.5	51.5	229.4
15*	4	6	41,13 SE	27.2	56.6	47.4	57.5	8.4	52.5	200.6
16*	4	5	16,10 NW	39.7	48.2	28.4	51.3	6.5	66.2	219.9
17*	3	11	32,8 SE	7.9	49.5	17.3	52.2	5.6	75.4	225.9
18*	4	12	45,14 SE	2.7	43.5	15.2	51.7	6.6	76.9	230.1

	N	Dg	Ig	α_{95}	Kg	Ds	Is	αs_{95}	Ks
SK I(mean of sites 03 to MV)	8	13.4	53.4	5.7	96	37	60.9	5.0	120
SKII(mean of sites MV to 18)	7	18.8	47.3	9.6	41.0	28	51	7.9	58.9
Overall mean	14	16.5	50.3	5.5	52.7	32.4	56.4	5.3	57.2

MV is the mean direction for the three volcanic sites. Nb is the number of blocks, n is the number of samples in each site, N is the number of sites, K the precision parameter, and α_{95} is the radius of the circle of 95% confidence. The latitudes (λ) and longitudes (ϕ) of VGPs are also given for each site. BA are the mean bedding attitudes for each site.

* The final mean only comprises the sites.

red sandstones; Figure 5): Hematite appears to be the main carrier, and af demagnetization was not effective. Thermal treatment isolates two or three components; in site 6, A appears between 40℗ and 50℗ C, B between 50℗ and 62℗ C, and high-temperature remagnetization occurs above (specimen 6-11A, Figure 5). Chemical demagnetization, which preferentially removes fine-grained hematite, first reveals component A after 3 days of leaching in 1 to 6 N HCl and then B with 5 days of leaching in 12 N acid (specimen 6-12A, Figure 5). Most samples of site 7 reveal only the component A carried by hematite (specimen 7-23A, Figure 5). Only a few specimens reveal component B between 40℗ and 66℗ C (specimen 7-11A), carried by hematite. Site 8 is important in that it is one of the very few to display component B (in the 40℗-62℗ C range) with both normal and reversed polarities (specimens 8-32B and 8-23A, Figure 6). The respective means (Table 1) are indeed antiparallel, which is consistent with a primary origin of the remanent magnetization. Despite a strong bias to the normal polarity, these results are in agreement with a magnetization acquired prior to the beginning of the long normal interval in the Cretaceous (i.e., ~110 m.y.).

Site 9 (Silla formation; red siltstones; Figure 6): The red siltstone layers in which the site is located are interbedded within the Silla conglomerate. Specimen 9-23B (Figure 6) is a typical example of the relative intensities of remanences A and B: A corresponds to more than 90% of the NRM, and B appears only above 65℗ C, carried by hematite. Despite its small size, the direction of B can be determined with good accuracy (Table 1).

Sites 11-13 (Hagbong volcanics; Figure 6): These sites are located in diabase flows interlayered within the Silla and Haman formations. We have taken for the bedding attitudes that of sedimentary rocks situated above or below the different flows. Af treatment reveals component A (specimen 11-32B, Figure 6), whereas B is carried by and 67℗ C in the thermal diagrams (specimen 13-31B, Figure 6). Individual site directions have very small variance in agreement with the volcanic nature of the rocks (Table 1), and they have been combined in an attempt to average out secular variation.

Sites 14 and 15 (Haman formation; red shales; Figure 6): Site 14 displays the two components A and B, with both polarities (specimens 14-51B, Figure 6). The B averages are indeed reasonably antiparallel (Table 1); the average inclination is shallower than in other sites, both in geo-

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

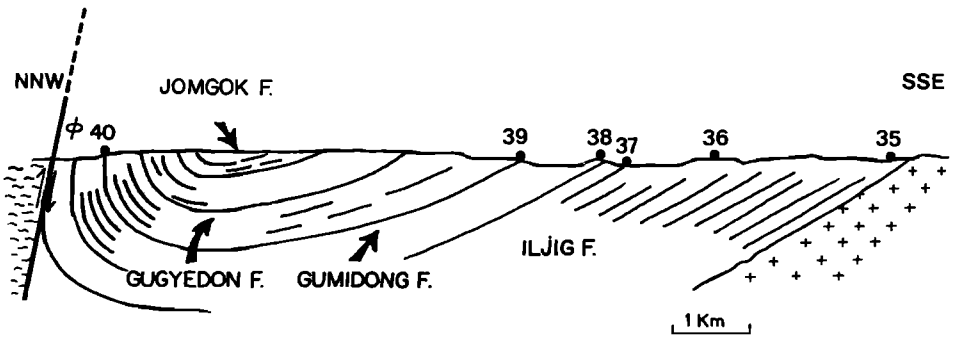


Fig. 7. Cross section southeast of Andong in the Dopyeong block and location of paleomagnetic sampling sites.

graphic and stratigraphic coordinates, particularly for the reversed specimens. This may be due to insufficient removal of a recent normal remagnetization or to errors in the determination of the local dip of the shales. Site 15 is very similar to others already encountered, such as 9, and a weak component B clearly emerges from a much larger A above 640°C (specimen 15-12A, Figure 6).

Site 16-18 (Banyaweol and Songnaedong formations; mudstones and red shales; Figure 6): These sites were sampled in the slightly more folded southeastern termination of the basin and provide a welcome opportunity for a fold test. In all sites, A and B are identified, below and above 500° to 550°C, respectively (specimens 16-21A and 17-11A; Figure 6). Component A is very close to the present dipole field before bedding correction ($D = 1°$, $I = 51°$, $\alpha_{95} = 18°$) and also after bedding correction ($D = 4°$, $I = 55°$, $\alpha_{95} = 9$). Although the bedding correction improves the cluster of the directions, the fold test is negative at the 95% level (K_s/K_g = 3.6). Component B is well determined and passes a positive fold test at the 95% probability level (K ratio of 9.9 with N = 3 sites, Table 1 [see McElhinny, 1964]). Although unfortunately based on only three sites, this is the only fold test available for the Gyeongsang basin.

Daegu section (results). The medium unblocking temperature component A is well determined in all sites. With a mean direction for 14 sites at $D = 7°$, $I = 55°$ ($k = 51$, $\alpha_{95} = 6°$) before structural correction, this direction is significantly different, at the 95% probability level, from either the recent dipole field direction ($D = 0°$, $I = 59°$) or the present field direction ($D = -6.5°$, $I = 50°$) in South Korea. After bedding correction the direction becomes $D = 24°$, $I = 62$ ($K = 30$, $\alpha = 8$), and the scatter increases slightly. Thus component A is likely to be a postfolding remagnetization, and, although it could well be of very recent origin, it may also be an older remagnetization.

The site mean directions for the higher-temperature component B are listed in Table 1, both before and after correction for bedding attitudes. The directions are generally consistent and show a significant eastward declination in geographic coordinates. Although an overall fold test is not available, the presence of a local fold test at sites 16-18 and of antiparallel directions at sites 8 and 14 and the consistency of directions in various rock types with various carriers (magnetite being a minor carrier, some forms of ilmeno-hematite being the major carrier) all argue that remanence B is primary or at least a chemical remanent magnetization acquired during diagenesis shortly after deposition. The mean direction prior to bedding correction ($D = 16.5°$, $I = 50°$, $\alpha_{95} = 5.5°$) is significantly different from recent field directions in South Korea.

We have attempted in Table 1 to separately average the sites from the lower (3-13) and upper (11-18) parts of the series. These means are labeled SK I and SK II in Tables 1 and 3. They would indicate a decrease in declination ($37°-28°$) and in inclination ($61°-51°$), although this is barely resolvable with α_{95} values of $9°$ and $8°$, respectively.

The final corrected mean direction is $D = 32°$, $I = 56°$ ($\alpha_{95} = 9°$) in stratigraphic coordinates. It should be recalled from the vector demagnetization diagrams of Figures 5 and 6, that the component B may not have been completely separated from a remaining influence of component A (compare for instance specimens 3-11A and 3-32A in Figure 5). Thus both declination and inclination could sometimes have been underestimated.

Andong section (measurements). In order to extend the results from the Daegu section, another section was sampled near Andong, about 120 km to the NNE, in the Dopyeong part of the Gyeongsang basin (Figure 3). The six sites which provided reliable results (over seven sampled sites) span a 15-km syncline (Figure 7) consisting of continental sediments lying unconformably over Precambrian gneisses. The formations have been stratigraphically correlated with the Chilgok sediments in the Nagdong trough [Chang, 1975]. However, we have already pointed out the difficulty in establishing accurate correlations in such paleoenvironments with high-energy continental sedimentation. The overturned northern flank of the syncline abuts a reverse fault (the Andong fault) which provided site 40 upon which a fold test was based. The demagnetization diagrams for sites 35-40 (Figure 8) are quite similar to those from the Daegu section and display the same three components. Details are given below, and the mean directions for the high-temperature component B are listed in Table 2.

Site 35 (specimen 35-12A, Figure 8): Only component A, probably carried by magnetite, is

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

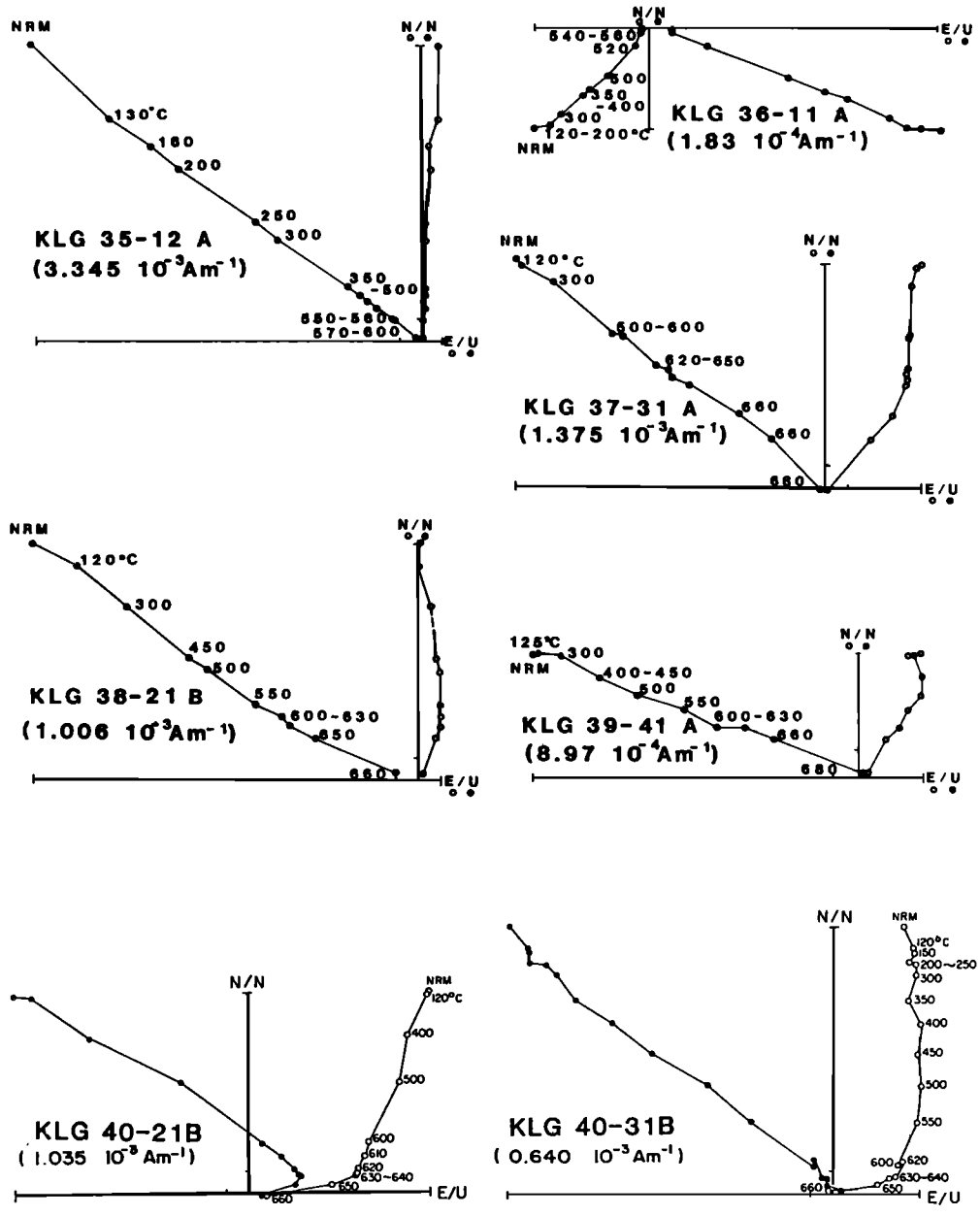


Fig. 8. Vector demagnetization diagrams of specimens from the Andong section. Same as Figure 5.

found in this site. The resulting mean direction D = 1°, I = 59° (α_{95} = 8°) is virtually identical with the present dipole field direction and is likely to be a recent remagnetization.

Site 36 (specimen 36-11A, Figure 8): Component B is found between 300° and 560°C and is reversed. Behavior is reminiscent of that in sites 8 and 14 (see Figure 6), with somewhat lower unblocking temperatures.

Sites 37 to 39 (specimens 37-31A, 38-21B and 39-41A, Figure 8): In these three sites, both components A and B are clearly identified, with component B appearing usually above 600°C, sometimes only above 660°C.

Site 40 (specimens 40-21B and 40-31B, Figure 8): This is a key site in several respects. At temperatures below 600°C a clear component is isolated, which we would like to associate with component A (e.g., specimen 40-31A between 250° and 600°C). However, in several specimens the declination in that temperature range is significantly rotated to the east (e.g. specimen 40-21B between 120° and 620°C). The mean for this component A' in the site is D = 15°, I = 57° (α_{95} = 3°) in geographic coordinates and thus significantly different from a recent field direction. The high-temperature component (640° - 660°C, shortly to be identified with B) has a large eastern declination and a scattered reversed inclination, going from quite steep upward (specimen 40-21B) to very shallow (specimen 40-31A), with intermediate cases. Individual specimen directions in geographic coordinates are found to lie in a plane containing the recent

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Directions of the High-Temperature Components for the Andong Section

Site	Nb	n	BA	Dg	Ig	$(\alpha_{95})_g$	Ds	Is	$(\alpha_{95})_s$	λ(N)	φ(E)
36	3	2	101,20 NW	225	-63	3	212	-56	2.8	64.4	209.0
37	4	20	122,16 NW	43	55.5		24.1	48.7	3.6	68.7	230.6
38	3	13	116,12 NW	21.8	66.1		7.3	56.9	2.7	84.1	206.7
39	3	10	157,3 SW	14.3	64.2	6.5	9	65.9	6.5	76.5	155.5
40	3	10	45,86 NE	76.5	-13.6		181.5	-54.8	6.7	88.3	261.4

	n	Dg	Ig	α_{g95}	Kg	Ds	Is	α_{s95}	Ks
SK III (overall mean)	5	0.9	70.9	44.2	3.9	15.4	57	8.9	74.4

See Table 1 footnotes.

field direction. Thus the directions isolated at very high temperatures are likely to retain a varying amount of recent remagnetization. The mean of 10 specimens displaying those directions farthest away from the present field are likely to be the least contaminated and are used to compute the mean of Table 2. If some amount of recent remagnetization would remain, this would imply that the corresponding declinations are underestimated and the inclinations areoverestimated (in stratigraphic coordinates).

Andong section (results). The low- to middle-temperature component from the six sites gives a mean direction D = 9°, I = 59° (α_{95} = 6°). This includes the present dipole field direction within statistical accuracy and fails the fold test at the 99% probability level (with Fisher's precision parameter k dropping from 145 to 4). It is also statistically not distinguishable from the corresponding component A from the Daegu section, with which it can be identified. This is a postfolding, possibly recent, remagnetization.

On the other hand, the very high temperature component provides a positive fold test which is significant at the 99% probability level (k ratio increasing from 4 to 74, see Table 2 [McElhinny, 1964]. The mean direction prior to bedding correction has a large error and unlikely steep inclination, whereas after correction it becomes D = 1°, I = 57° (α_{95} = 9°), labeled as SK III in Table 2. The fold test (even though it relies on only one site) and the fact that two out of six sites show a reversed component argues for a prefolding, possibly primary magnetization. In view of the fact that declinations for the high-temperature component could be underestimated due to remaining overprints, the mean direction for the component B of the Andong section compares favorably with the component B of the Daegu section (Figure 9): the inclinations are the same and the declination difference is 16° (±11°). Also, it is virtually identical with the postfolding remagnetization A' at site 40.

4. Discussion of South Korean Paleomagnetic Results

The main result that emerges from this study is that a high unblocking temperature component, often uncovered only above 600°-650°C and amounting to a small fraction of the total NRM, can be identified in most Lower Cretaceous continental rocks of the Gyeongsang basin (18 sites spread over an ~150 km area). Two local fold tests and the presence of four sites with reversed polarity suggest that this characteristic remanent magnetization is primary, or at least a chemical magnetization acquired shortly after

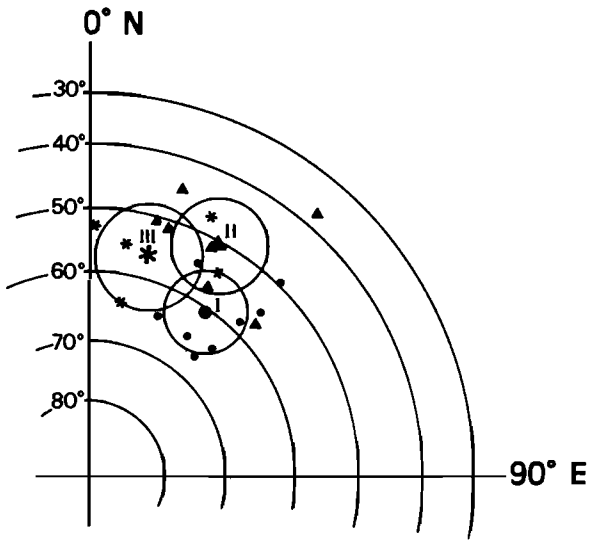


Fig. 9. Equal-area projection of characteristic mean directions of all South Korean sites after bedding correction. Solid dots are the directions of sites 3-MV, triangles are for sites 14-18 and stars for sites 36-40. The mean for each group of sites (see text and Tables 1 and 2) with their 95% confidence limits are also shown as larger symbols.

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Virtual Geomagnetic Poles for Southeastern Asia

	Source	N	VGP λ(N)	VGP ϕ(E)	A_{95} dp/dm	expected directions D	expected directions λ
Tertiary pole for China							
China, Shandong 18 m.y.	Lin [1984]	1	76	178	6		
Cretaceous poles for southeastern Asia							
Sites							
South Korea I*	This paper	8	60.8	194.2	6.5	37.1	42.7
South Korea II*	This paper	7	66.4	219.3	9.2	28.3	32.5
South Korea III*	This paper	5	77.9	207.1	11.6	15.0	37.6
South Korea I+II+III*		19	67.6	205.1	5.8	28.1	38.1
North Korea	Gurarii et al. [1966]	1	69.0	182	3	24.5	46.3
Shandong N China	Lin [1984]	5	69	200	12.0	26.3	40.0
Zhejiang S China	Lin [1986]	7	76.2	225.7	4.8	16.5	33.3
N+S China*	Lin [1984]	12	74.3	211.4	6.3	19.3	36.5
Zhejiang	Kent et al. [1986]	?	76	173	10	13.8	45.3
North Japan Kitakami	Sasajima [1981]	6	52.0	520	9.0	-47.2	35.1
North Japan Miyako	Tosha [1982,1983]	6	79.7	185.2	7.3/3.5	11.4	41.2
South Japan	Sasajima [1981]	10	45.9	196	6.0	56.5	39.5
Mean Lower Cretaceous Pole for South Asia		31	70.2	206.3	4.1	24.7	37.8
Upper/Middle Jurassic poles for China							
Shandong N China	Lin [1984]	14	71.5	216.5	6.5		
Zhejiang S China	Lin [1984]	9	73.3	210.6	10.0		
N+S China	Lin [1984]	23	72.2	214.3	5.3		

South Korea I and II are from the Daegu section, III from the Andong section. λ and ϕ are the VGP latitude and longitude, A_{95} is the radius of the 95% cone of confidence. D, λ are expected declinations and paleolatitudes reduced to Daegu (36° N, 129° E).

* Computed as the mean of site VGPs.

deposition. The magnetization is carried most of the time by hematite but also sometimes by magnetite. The mean site directions for all sites in the basin are shown in Figure 9, together with the means and confidence intervals for the three groups of sites (SK I for the lower part of the Daegu section, SK II for the upper part, and SK III for the Andong section). Although the three confidence circles do intersect, the means remain outside of these intersections, and the differences between these means may have some significance. For instance, the inclination difference between SK I and SK II might be interpreted in terms of a slight southward shift of the Gyeongsang basin during deposition. Since the Andong section is correlated with the middle of the Daegu section, the SK III direction might be taken to imply a slight differential rotation between the two subunits of the Gyeongsang basin. Alternatively, in view of the difficulty in establishing correlations in the basin, SK III might reflect a stratigraphic age younger than generally accepted. These differences are indeed slight, and the correlation problems may be aggravated by the possibility of remaining overprints (for instance in SK III), which lead to an underestimate of declination. Although this discussion should be kept in mind, one may prefer to conservatively average all site-mean directions together, yielding a final mean direction of D = 29.5°, I = 57° (α_{95} = 5°), which is considered as representative of South Korea for the period 120 m.y. (±10 m.y.). This corresponds to a pole at 66° N, 206° E (dp = 5°, dm = 7°). Alternatively, the VGPs for individual sites can be averaged, leading to a mean pole at 68° N, 205° E (A_{95} = 6°, Table 3).

We next compare our data with earlier studies available for South Korea. Otofuji et al. [1983] obtained data from five sites in the Gyeongsang basin. Four of the sites were demagnetized at 20-30 mT, and one was thermally demagnetized up to 150° C. The above discussion shows that these demagnetization levels are not sufficient to

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

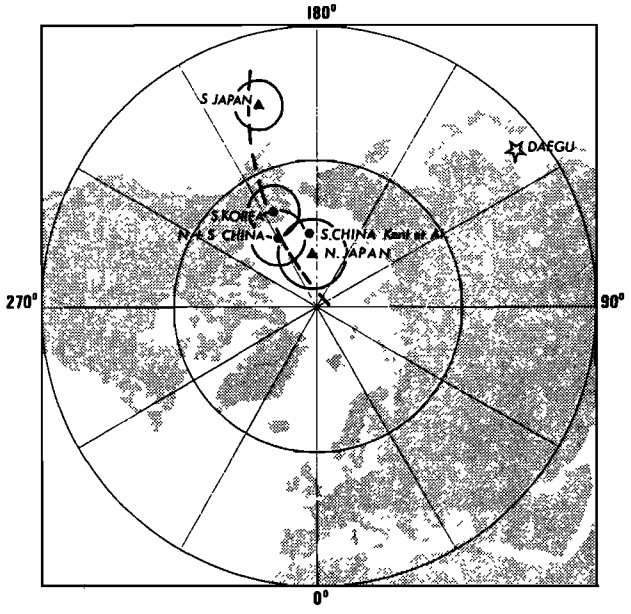


Fig. 10. Equal-area projection of the Lower Cretaceous poles from China, Korea, South Japan and North Japan in the hypothesis of a primary magnetization of Miyako sediments (see text). All blocks occupied the same relative position in term of latitudes. The common small circle is centered on the Japan Sea.

isolate the high-temperature component B. Indeed, the final mean direction obtained by Otofuji et al. [1983] is D = 2°, I = 62° (α_{95} = 8°) and is given after bedding correction. Should the post folding remagnetization A from the Daegu section be corrected for bedding, a direction D = 2°, I = 62° (α_{95} = 8°), identical to the one found by Otofuji et al. [1983] would be obtained. However, we have shown that this correction should not be applied because A is a remagnetization.

Another recent study is that of 80- to 100-m.y.-old granites by Ito and Tokieda [1980]. Unfortunately, no stability test is available: All polarities are normal, which would agree with the ages being within the long normal interval of the Cretaceous, and no structural test can be performed on plutonic rock bodies. The mean direction found by Ito and Tokieda [1980], D = 3.5°, I = 57° (α_{95} = 8°), is virtually identical with the present dipolar field direction in South Korea, a recent overprint being the most likely interpretation.

5. Comparison With Data From Neighbouring Blocks

China. The neighbouring North and South China blocks were until recently almost devoid of reliable paleomagnetic data. The situation has been changed by the publication of Lin's [1984] and Lin et al.'s [1985] data, which span the two blocks for the period going from the Cambrian to the Tertiary. This important set of data deserves a detailed discussion. The conclusions based on the reanalysis of these data will be seen to vary at times with the inferences of Lin et al. [1985] (see also Courtillot and Besse [1986]).

Averaging middle to upper Jurassic poles from Lin et al. [1985] (see Table 3) separately for the two blocks, we find that the error on the SCB pole is rather large (α_{95} = 10°) and the corresponding confidence interval entirely encloses that of the NCB. This is an agreement with the pre-Jurassic suturing of the two blocks. A recent geological study of the Qinling mountains, which separate the North China block (NCB) from the South China block (SCB), by Mattauer et al. [1985] concludes that the two cratons had actually collided during the Carboniferous and that lower Mesozoic deformation was an intracontinental event. Lin et al.'s [1985] paleomagnetic data are in agreement with this scheme, and we can combine them to propose a middle/upper Jurassic pole for China as a whole at 72° N, 214° E (A_{95} = 5°) (excluding one anomalous JM pole from the NCB: pole NC 30 in Lin's [1984] list).

The lower Cretaceous poles for the two blocks are also not statistically distinct. They can be combined, confirming that the two Chinese blocks remained as a coherent unit during that period: The resulting mean pole at 74° N, 211° E (A_{95} = 6°, Table 3) is indistinguishable from the mean Jurassic pole. We are thus led to the conclusion [Courtillot and Besse, 1986] that the Chinese blocks did not suffer significant internal deformation and remained fixed with respect to the earth's spin axis for a period extending at least from middle Jurassic to lower Cretaceous (included). All these results can in turn be combined to determine a mean pole at 74° N, 214° E (A_{95} = 4°).

We can see from Figure 10 that the lower Cretaceous poles for South Korea and China are not significantly different. The expected declination and paleolatitude at a reference point near Daegu (36° N, 129° E) are (D = 28°, λ = 38°) and (D = 19°, λ = 36°), respectively. The paleolatitude difference is 2° ± 12°, and the declination difference 9° ± 11°. This confirms that within available paleomagnetic accuracy, the Korean peninsula can indeed be considered as a part of the Chinese blocks (in the lower Cretaceous).

The upper Cretaceous poles of Lin et al. [1985] actually correspond to Korean data. The NCB result is the single site pole from North Korea of Gurarii et al. [1966], and the SCB result is the Otofuji et al. [1983] pole for the Gyeongsang basin which we have already suggested to be an overprint.

The only published pole for the upper Cretaceous of the SCB is that of Kent et al. [1986] at 76° N, 173° E (A_{95} = 10°), which is significantly different from the lower Cretaceous pole for the same block.

Lin [1984] pointed out that the Tertiary pole for the SCB comes from only three lava flows in a single site. Reversed directions for one site [Lin, 1984, p. 167] suffer from overprints and insufficient cleaning and one out of three K/Ar ages is clearly anomalous [Lin, 1984, p. 165]. The resulting mean Tertiary pole (74° N, 36° E), which is discordant, is therefore doubtful and should be discarded until further data are available. On the other hand, the Tertiary pole for

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the NCB is close to the Jurassic and lower Cretaceous poles but again relies on four lava flows from a single site (Table 3).

Japan. Japan is traditionally separated in two distinct domains, the North and South Japan blocks, usually on either side of the Fossa Magna (e.g., Kawai et al., [1961] and FM on Figure 1). Recent work has emphasized the tectonic significance of the NNW trending Tanakura and Itoigawa-Shizuoka tectonic zones [e.g., Faure et al., 1986].

North Japan: The abundant paleomagnetic work carried out over the last decades in Japan has been summarized by Sasajima [1981]. For northern Japan, Sasajima lists six "reliable" poles based on Barremian to Albian (107-121 m.y.) plutonic rocks from the Kitakami massif in the northeastern part of Honshu. These yield a mean pole at 52°N, 52°E (A_{95} = 9°). Unfortunately, no stability test is available, and of course, no structural tests or corrections are possible in these plutonic bodies. More recently, Tosha [1982, 1983] carried out an extensive analysis of Devonian to Tertiary sediments in North Honshu, particularly around the Kitakami massif. Tosha [1983] clearly demonstrated through negative fold tests on Devonian to Jurassic data that the whole area suffered widespread (virtually total) remagnetization after the Jurassic. From Tosha's [1983] data, one can derive a mean direction D = 32°, I = 50° (α_{95} = 8°) corresponding to a pole at 62°N, 44°E. This is rather close to Sasajima's pole for the lower Cretaceous granites. Tosha [1983] considers that the Kitakami granite magnetization in geographic coordinates provides a correct estimate of the lower Cretaceous paleomagnetic direction for this area and that no major tilting of the plutons occurred after their emplacement. Tosha [1982, 1983] also provides paleomagnetic data for siltstones and sandstones from the late Aptian to Albian Miyako Group, which occurs as a tilted (but hardly folded) sequence unconformably overlying the granites and earlier sediments. Six sites provide an in situ direction D = 322°, I = 56° (α_{95} = 7°), which becomes D = 10°, I = 65° (α_{95} = 8°) if corrected for bedding. Although the series is essentially monoclinal and does not provide a fold test, Tosha proposes that the Miyako Group magnetization is primary. This is based in particular on a positive conglomerate test on the lowermost part of the formation which would suggest that the area was not subjected to post-Albian remagnetization. Tosha's interpretation leads to a major directional change between the Aptian (emplacement of the granites) and the Albian (deposition of the unconformable Miyako series). This results in a major loop in his proposed apparent pole wander path APWP of northeast Japan between 100 and 125 m.y. [Tosha, 1983, Figure 7.1, p. 73]. However, the Miyako Group paleomagnetic direction prior to tilting is remarkably close both to the Kitakami granite and to the Devonian to Jurassic remagnetization directions, suggesting either one of two possibilities:

1. The Kitakami massif could have suffered major tilting prior to deposition of the Miyako sediments. In that case, Sasajima's [1981] mean direction could not be considered as a valid indication of the lower Cretaceous magnetic field in NE Japan, and Tosha's [1982, 1983] direction would be the only valid one, leading to an Albian-Aptian pole at 80°N, 185°E (A_{95} = 5°). This pole can be directly compared to the mean South Korean pole (Figure 10). The two 95% confidence circle are tangent, and the poles are roughly at the same distance from the sampling sites. This translates into a paleolatitude difference of 3° (±8°) which is not significantly different from zero, and a declination difference of 17° (± 10°) which is significant. This would correspond to a ~ 20° counterclockwise rotation of northeast Japan with respect to Korea (and the rest of the mainland) since upper Cretaceous time.

2. However, the conglomerate test on the Miyako Group is based on four pebbles only and may not be very strong evidence. Hence both the Miyako and the Kitakami massif could have suffered widespread post-Cretaceous remagnetization (M. G. Moreau et al., manuscript in preparation, 1986). In that case, we would not have any pre-Cenozoic information on the position of North Japan. The event which is most likely to have led to such a remagnetization is the opening of the Japan Sea, which is generally supposed to be an Oligo-Miocene event. In that case, the remagnetization direction compared with the direction expected for Eurasia in the Miocene [Achache et al, 1983] would imply a 35° ± 10° counterclockwise rotation.

South Japan: An abundant, although uneven, set of data is available for the Southwest part of Honshu. Sasajima [1981] summarizes results from 10 studies of mostly volcanic and some sedimentary rocks and determines a consistent pole at 46°N, 196°E (A_{95} = 6°). The comparison with the South Korean pole demonstrates no latitude difference (1° ± 8°) and a significant declination difference (28° ± 11°), despite a rather large uncertainty. The declination difference corresponds to a post lower Cretaceous rotation of Japan of 28° in a clockwise sense with respect to Korea and the mainland. This is in the same sense and has the same magnitude as proposed by Sasajima [1981] but is significantly smaller than the 54° proposed by Otofuji and Matsuda [1983] to have occurred around 10-20 m.y. or 45° proposed by Torii [1983] from a study of Middle Miocene rocks.

Comparing the results for South and North Japan, we see that a 55° bending would be implied in the hypothesis of primary Cretaceous magnetizations in the Kitakami massif. A larger value of 70° is obtained if the alternate hypothesis of an Oligo-Miocene remagnetization is accepted. The latter figure is close to values proposed in previous studies [Sasajima, 1981; Otofuji and Matsuda, 1983].

Eurasia and Siberia: The lack of reliable data of Cretaceous and Cenozoic age from the Siberian block, and also from the rest of Eurasia, is a well-recognized problem. Many upper Mesozoic poles from Siberia include data from orogenic areas and also data that have not been adequately thermally demagnetized. Published directions may thus be significantly biased by remaining later overprints. We have selected in the recent compilation of Khramov et al. [1981] all the poles with a lower Cretaceous age that

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Virtual Geomagnetic Poles for the Eurasian Plate

Age	Eurasian Poles References	λ(N)	ϕ(E)	A_{95}	D	λ
Jl*	Westphal et al.[1986]	65	122	4	-6.0	60.7
Jm*	Westphal et al.[1986]	69	158	7	16.8	53.4
Jl-Jm	Irving [1977]	65	139	9	8.5	60.4
Ju	Irving [1977]	76	162	9	11.2	47.3
Kl	This paper	72	164	5	15.9	49.8
Kl	Irving [1977]	70	153	6	13.6	53.6
Kl	Westphal et al.[1986]	70	166	4	18.9	50.7
Kl*	Van der Voo and French[1974]	73	211	14	21.0	36.5
Ju-Kl*	Westphal et al.[1986]	66	194	6	30.0	42.5
Kl*	Westphal et al.[1986]	68	208	6	27.4	37.1
Km*	Westphal et al.[1986]	69	211	7	26.1	36.1
Ku*	Westphal et al.[1986]	73	212	8	21.1	36.2
65m.y.	Westphal et al.[1986]	80	186	6	11.1	40.9
Ku	Irving [1977]	69	164	6	19.4	68.5
Pa/E	Achache et al.[1983]	75	168	4.3	14.0	46.8
E/O	Achache et al.[1983]	75.4	155.6	4.4	9.8	48.7
M	Achache et al.[1983]	80	148	4	4.6	45.4
M/P	Achache et al.[1983]	83.5	167	4.9	5.3	41.0

See Table 3 footnotes.
* Transferred data.

were remote from tectonically disturbed areas (Table 4). The resulting mean includes 14 studies and lies at 72° N, 164° E (A_{95} = 9°), actually close to the smoothed estimate of Irving [1977] (Table 4) at 70° N, 159° E (A_{95} = 6°). Westphal et al.'s [1986] independent reanalysis of Eurasian data yields again a nearly identical pole for the lower Cretaceous at 70° N, 166° E (A_{95} = 4°), based on nine studies. These lower Cretaceous poles are significantly different from the corresponding pole for the Sino-Korean block (Figure 11). The expected declination and paleolatitude at our reference point near Daegu is D = 16°, λ = 50°, which would imply a declination difference of 9° (± 9°) and a latitude difference of 12° (± 6°). The corresponding figures using Irving's [1977] pole are ΔD = 11° (± 11°) and Δλ = 16° (± 7°).

An alternate way to determine a lower Cretaceous pole for Eurasia is to transfer data from another, supposedly better known plate (for instance North or South America), using kinematic models based on the identification of oceanic magnetic anomalies. This approach is of course very sensitive to inaccuracies in the original pole determinations and to errors in position and timing of the Euler rotation poles deduced from the oceanic reconstructions. Using this technique, Van der Voo and French [1974] obtained a "transferred" lower Cretaceous pole for Eurasia at 73° N, 211° E, with a rather large uncertainty of 14°. More recently, Westphal et al. [1986] transferred the South American and recent North American poles of Harrison and Lindh [1982] to Eurasia: for the upper lower Cretaceous, i.e., the age of the Gyeongsang series, they find a pole at 69° N, 211° E (A_{95} = 7°). This is statistically identical with the pole which we obtain for South Asia (Korea and China, Table 3) at 70° N, 206° E (A_{95} = 4°). A recent study of Valanginian and Hauterivian samples from the south of France (J. Besse and T. Boisseau, manuscript in preparation, 1986) yields a very similar pole at 67° N, 220° E (A_{95} = 9°).

We are thus led to two conflicting possibilities. Either the "classical" determinations from the Eurasia/Siberia craton are acceptable and then the Korea/China block has a significantly

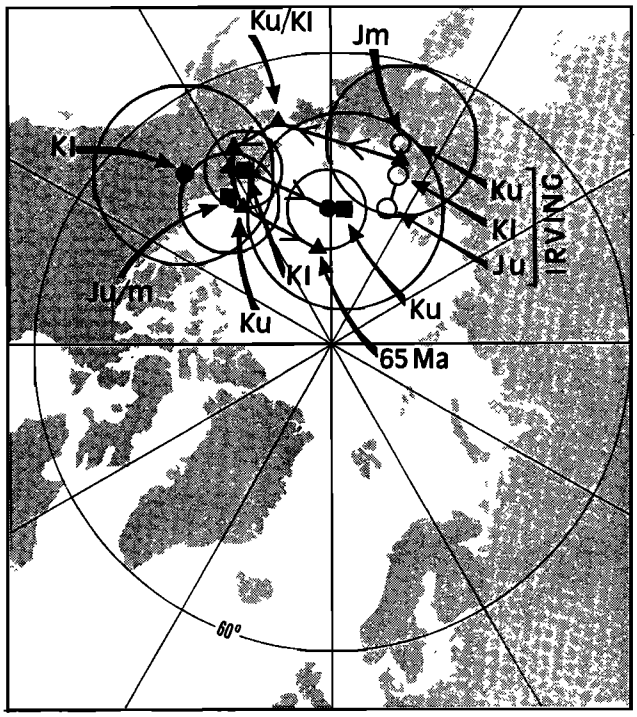


Fig. 11. Equal-area projection of Jurassic and Cretaceous VGPs for the Sino Korean block (solid squares) and Europe (solid circles). Triangles represent the transfered APWP from Westphal et al. [1986]; open circles represent the on-land data APWP (see text).

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

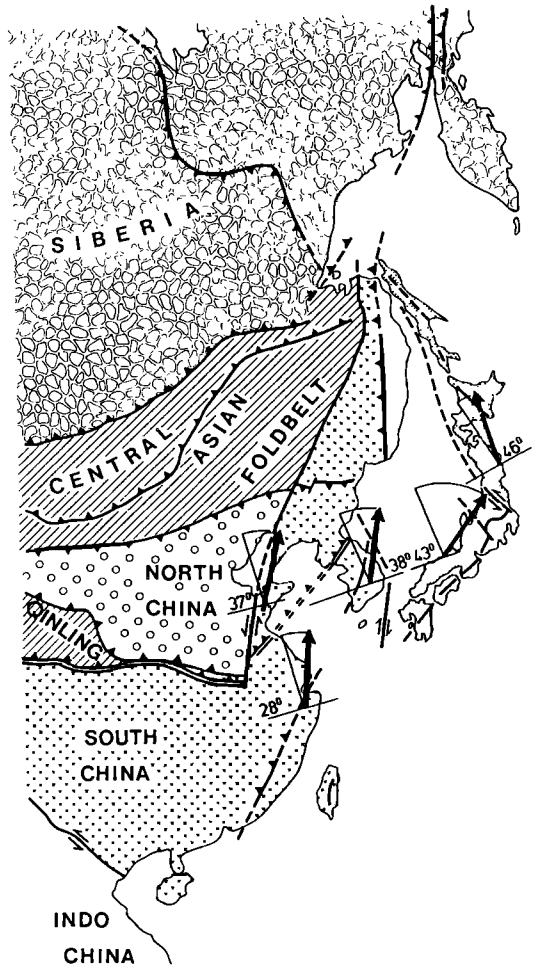


Fig. 12. Sketch map of Southeastern Asia indicating major tectonicunits. The arrows represent the absolute post-Cretaceous rotation for each block, the figures nearby are the Cretaceous paleolatitudes. Black barbed lines, continental thrusts or ancient suture zones. Heavy lines, main strike-slip faults. Dashed lines; hypothetical strike-slip faults.

distinct lower Cretaceous pole, which entails a paleolatitude discrepancy of 12° (±6°); or the "classical" determinations are in error, the transferred poles are to be preferred, and then the Korea/China block already formed a coherent unit with the Eurasia/Siberia craton.

We note that upper Cretaceous pole determinations are more numerous than lower Cretaceous ones and are in good agreement. For instance, the pole of Kent et al. [1986] for South China at 76° N, 173° E (A_{95} = 10°) and the European pole of Heller and Channel [1979] at 76° N, 181° E (A_{95} = 4°) are very close to each other. However, they are some distance from the corresponding transferred pole of Westphal et al. [1986] at 73° N, 212° E (A_{95} = 8°).

The Eurasian transferred data, the lower Cretaceous data for Korea, China, and France, the upper Cretaceous data for South China and Germany taken all together would imply, under the hypothesis that they formed a stable coherent unit from the lower Cretaceous onward, a loop in the APWP of Eurasia which is absent from Irving's [1977] compilation. More data are clearly required to refine and confirm this picture.

6. Discussion and Conclusions

The new paleomagnetic data from the lower Cretaceous of southern Korea and the estimates from North and South China and Japan (Figure 12) discussed in section 5 allow a number of inferences to be made concerning the geodynamic evolution of eastern Asia in the last 120–130 m.y.

The first major conclusion to emerge from this data set is the fact that all these parts of eastern Asia occupied the same relative position in terms of latitudes in the lower Cretaceous: This is clear from Figure 10 where the poles for North and South Korea, North and South China, South Japan and North Japan (in the hypothesis of a primary magnetization for Miyako sediments) all lie on a small circle centered roughly on the Japan Sea. For instance, the mean paleolatitudes predicted for Daegu (Table 3) using the various poles average to 39° N, with a variance of only 2°. Moreover, this small circle passes close to the present north pole, meaning that the lower Cretaceous and present latitudes are the same (36° N at Daegu presently). The differences are in terms of relative rotations. The rotation between South Korea and China may not be significant (9° ± 11°). Much of the error is due to the spread in declinations between data from the Daegu and Andong sections (Figure 9). Since remaining recent overprints may have led to underestimates of declination of the characteristic remanent magnetization, some (minor) amount of rotation is a possibility. Notwithstanding this minor rotation, and the larger rotations of Japan related to the Cenozoic opening of the Japan Sea, we find that Korea, China, and Japan formed a coherent (isolatitudinal) block in the lower Cretaceous.

We point out (see also Courtillot and Besse [1986]) that all available paleomagnetic sites from the North and South China blocks for the period from upper Jurassic to Tertiary [Lin et al., 1985] come from two very small areas (only 100–150 km wide) near Hangzhou in Zhejiang province and Jinan in Shandong province, respectively (Figure 1), on the eastern borders of the two blocks. Thus the interior of the North China block, west of the Tanlu fault, is still devoid of paleomagnetic data. The relative motion of the South China/Korea/Japan block with respect to Eurasia/Siberia since the lower Cretaceous depends on which one of the two conflicting possibilities for Eurasia/Siberia is right. Should one accept the pole based on Siberian data (Irving [1977], Westphal et al. [1986] or this paper), a 12° (±6°) or 1300 km (±700 km) latitudinal convergence between South China and Siberia must have taken place in the last 100 m.y. This motion could have occurred either between South and North China or between the combined Chinese blocks and Siberia. In the latter case, major tectonic deformation of upper Cretaceous or Cenozoic age would be implied in the Central Asian foldbelt. However, it is generally accepted that major deformation in this foldbelt was terminated long before the Creta-

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ceous. Tapponnier and Molnar [1977] conclude from a detailed study of active faulting and Cenozoic tectonics that the motion of China with respect to Eurasia should have taken place in an EW left-lateral sense, with a total Cenozoic amount not exceeding 300 km. An other possibility would be motion across the Qinling range and along the Tanlu fault zone. Although the sense and amount of motion deduced from the paleomagnetic data are in good agreement with some field data [Xu, 1980; Lu et al., 1983], recent observations across the Qinling range [Mattauer et al., 1985] seem to preclude any large-scale convergence since the lower Mesozoic. Post-Jurassic motion is observed to be EW strike slip not exceeding 200 km [Peltzer et al, 1985].

This discussion favors a model in which North and South China were already accreted into a single continental landmass in the lower Cretaceous. In that case, our South Asia pole (Table 3) would supercede earlier Siberian data, which would be in good agreement both with the European data of J. Besse and T. Boisseau (manuscript in preparation, 1986) and with the transferred pole of Westphal et al. [1986]. On the other hand, the transferred Jurassic poles are quite distinct from the Chinese poles [Lin et al., 1985; Courtillot and Besse, 1986], which would seem to imply accretion of China against Siberia sometimes in the Jurassic. However, more paleomagnetic and geological data are required before this suggestion can be seriously tested. The upper Cretaceous data of South China [Kent et al., 1986] and Germany [Heller and Channell, 1979] would imply a loop in the Cretaceous APWP of Eurasia. In that case, the similarity of the lower Cretaceous and present latitudes of the China/Korea/Japan blocks would not imply long-term latitudinal stability.

Acknowledgments. We wish to acknowledge help in the field from Eric Humler (geochronology), Eric Buffetaux and Jean Marcoux (paleontology and stratigraphy), and Lee Byung Joo, Michel Faure, Jean-Paul Cadet, and Laurent Jolivet (tectonics). We gratefully acknowledge practical assistance from the Korean Institute of Energy and Resources and its President Keung-Shik Park, and from the French Embassy in Seoul (Thierry Audric). Reviews by D. Kent and R. McCabe were very helpful. This work was partly supported by Centre National de la Recherche Scientifique - Institut National des Sciences de l'Univers - Action Spécifique Programmée "Blocs et Collisions." Institut de Physique du Globe Contribution 917.

References

Achache, J., V. Courtillot, and J. Besse, Paleomagnetic constraints on the Late Cretaceous and Cenozoic tectonics of southern Asia, Earth Planet. Sci. Lett., 63, 123-136, 1983.

Chang, K. H., Cretaceous stratigraphy of southeast Korea, J. Geol. Soc. Korea, 11, 123-136, 1975.

Chinese Academy of Sciences, Tectonic map of Asia, Editor in Chief Li C. Y., Chinese Geological Cartographic Printing House, Beijing, 1982.

Courtillot, V., and J. Besse, Mesozoic and Cenozoic evolution of the north and south China blocks, Nature, 320, 86-87, 1986.

Faure, M., F. Lalevée, Y. Gusokujima, J. T. Iiyama, and J. P. Cadet, The Pre-Cretaceous deep seated tectonics of the Abukuma Massif and its place in the structural framework of Japan, Earth Planet. Sci. Lett., 77, 384-398, 1986.

Gurarii, G. Z., P. N. Kropotkin, M. A. Pezner, R. V. Son and M. V. Trubikhin, Laboratory evaluation of the usefulness of North Korean sedimentary rocks for paleomagnetic studies, Izv. Acad. Sci. USSR Phys. Solid Earth, 11, 128-136, 1966.

Harrison, C. G. A., and T. Lindh, A polar wander curve for North America during the Mesozoic and Cenozoic, J. Geophys. Res., 87, 1903-1920, 1982.

Heller, F., and J. E. T. Channell, Paleomagnetism of Upper Cretaceous limestones from the Munster basin, J. Geophys., 46, 413-427, 1979.

Irving, E., Drift of the major continental blocks since the Devonian, Nature, 270, 304-309, 1977.

Ito, H., and K. Tokieda, An interpretation of paleomagnetic resultsfrom Cretaceous granites in South Korea, J. Geomag. Geoelectr., 32, 275-284, 1980.

Kawai, N., H. Ito and S. Kume, Deformation of the Japenese Islands as inferred from rock magnetism, Geophys. J. R. astr. Soc., 6, 124-130, 1961.

Kent, D. V., G. Xu, K. Huang, W. Zhang and N.D. Opdyke, Paleomagnetism of upper Cretaceous rocks from South China, Earth Planet. Sci. Lett., in press, 1986.

Khramov, A. N., G. N. Petrova, and D. M. Pechersky, Paleomagnetism of the Soviet Union, in Paleoreconstruction of the Continents, Geodyn. Ser., vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp.177-194, AGU, Washington, D. C., 1981.

Kim S. W., Petrology of the Late Cretaceous volcanic rocks in the Northern Yucheon basin: Implications for its tectonics, Ph.D. thesis, 133pp., Seoul Univ., Seoul, 1982.

Klimetz, M. P., Speculations on the Mesozoic plate tectonic evolution of eastern China, Tectonics, 2, 139-166, 1983.

Kobayashi, K., Spreading of the Sea of Japan and drift of the Japanese islands: A synthesis and speculation, Min. Geol. Tokyo Spec. Issue, 11, 23-26, 1983.

Kobayashi, K., and N. Isezaki, Magnetic anomalies in the Sea of Japan and the Shikoku basin: Possible tectonic implications, in The Geophysics of the Pacific Ocean Basin and Its Margin, Geophys. Monogr. Ser., vol. 19, edited by G. H. Sutton, M. H. Manghnani, and R. Mob, pp. 235-251, AGU, Washington, D.C., 1976.

Lee G., Paléomagnétisme des formations crétacées de la Corée du Sud: Contraintes géodynamiques, 3rd cycle Ph.D. thesis, 120pp., Univ. of Paris VII, Paris, 1984.

Lin, J., The apparent polar wander paths for the North and South China blocks, Ph.D. thesis, 248pp., Univ. of Calif., Santa Barbara, 1984.

Lin, J., M. Fuller, and W. Zhang, Preliminary

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

phanerozoic polar wanderpaths for the North China, Nature, 3313, 444-449, 1985.

Lu, H., H. Yu, Y. Ding, and Q. Zhang, Changing stress field in the middle segment of the Tanlu fault zone, eastern China, Tectonophysics, 98, 253-270, 1983.

Ma X., Q. Deng, Y. Wang, and H. Liu, Cenozoic graben systems in North China, Z. Geomorphol., 42, 99-116, 1982.

Mattauer, M., P. Matte, J. Malavieille, Z. Xu, Y. Lu, Y. Tang, and Z. M. Zhang, A section across the Paleozoic and Mesozoic Qinling belt (China): Tectonic evolution, plate tectonic implications, Nature, 317, 496-500, 1985.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc., 8, 338-340, 1964.

Molnar, P., and P. Tapponnier, Cenozoic tectonics of Asia: Effects of a continental collision, Science, 189, 419-426, 1975.

Otofuji, Y., and T. Matsuda, Paleomagnetic evidence for the clockwise rotation of southwest Japan, Earth Planet. Sci. Lett., 62, 349-359, 1983.

Otofuji, Y., Jin Y. O., T. Hirajima, Kyung D. M., and S. Sasajima, Paleomagnetism and age determination of Cretaceous rocks from Gyeongsang basin, Korean peninsula, in The Tectonic and Geologic Evolution of Southeast Asian Seas and Islands, Part 2, Geophys. Monogr. Ser., vol.27, edited by D. Hayes, pp.388-396, AGU Washington, D. C., 1983.

Peltzer, G., P. Tapponnier, Z. Zhang, and Z. Xu, Neogene and Quaternary faulting in and along the Qinling Shan, Nature, 317, 500-505, 1985.

Sang H. U., and Hee Y. C., Geology of Korea, 74pp., Korea Institute of Energy and Resources, Seoul, 1983.

Sasajima, S., Pre-Neogene Paleomagnetism of Japenese Islands (and vicinities), in Paleoreconstruction of the Continents, Geodyn. Ser., vol.2, edited by M. W. McElhinny and D. A. Valencio, AGU, Washington, D. C., pp.115-128, 1981.

Suh, H. G., and S. H. Choo, Intrusion ages of granites in Korea peninsula: Geological ore deposits, Rep. Geol. Surv. Korea, 14, 31-43, 1971.

Tapponnier, P., and P. Molnar, Active faulting and tectonics in China, J. Geophys. Res., 82, 2905-2930, 1977.

Tapponnier, P., G. Peltzer, and R. Armijo, On the mechanics of the collision between India and Asia, Collision Tectonics, edited by M. Coward and A. Ries, Geol. Soc. Am. Spec. Publ., 19, 115-157, 1986.

Torii, M., Paleomagnetic evidence for rapid clockwise rotation of Southwest Japan at Middle Miocene time, Rock Magn. Paleogeophys., 10, 77-79, 1983.

Tosha, T., Paleomagnetic Measurement of the Miyako group in the Kitakami massif, Rock Magn. Paleogeophys., 9, 56-60, 1982.

Tosha, T., Paleomagnetism of northeast Japan, Ph.D. thesis, 108pp., Univ. of Tokyo, Tokyo, 1983.

Van der Voo, R., and R. B. French, Apparent polar wandering path for the Atlantic bordering continents: Late Carboniferous to Eocene, Earth Sci. Rev., 10, 99-119, 1974.

Westphal, M., M. L. Bazhenov, J. P. Lauer, D. M. Pechersky, and J. C. Sibuet, Paleomagnetic implications of the evolution of the Tethys belt from the Atlantic Ocean to the Pamirs since the Triassic, Tectonophysics, 123, 37-82, 1986.

Xu, J., Horizontal displacement along the Tancheng-Lujiang fault zone and its geological significance (abstract), Int. Geol. Congr., 26 (1), 410, 1980.

Yu, C. J., The Tertiary fossil pollen and diatoms from Bukpyeong, Korea, Rep. Geol. Min. Geol. Surv. Korea, 13, 1971.

J. Besse, V. Courtillot, and Lee G., Laboratoire de Paléomagnétisme et Géodynamique, Institut de Physique du Globe de Paris, Université Paris 7, 75252 Paris Cedex 05, France.

R. Montigny, Institut de Physique du Globe de Strasbourg, Université Pasteur, 5 rue René Descartes, 67084 Strasbourg, France.

(Received July 17, 1985;
revised October 13, 1986;
accepted November 7, 1986.)

21562202b, 1987, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB092iB05p03580 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License